Manufacturing features

National 5 · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ identify split lines, flashing, ejector marks, draft angles and injection points
- ☐ explain the purpose of bosses, location pins, ribs and webs

I DO — Worked example

Read how a full-mark answer is built

A TV remote control casing is injection moulded in two parts.

1 Describe the purpose of a boss and of a rib on the inside of the casing. (2 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Boss: takes a screw that holds the two halves together, and strengthens the fixing point
- ✓ Rib: adds strength and rigidity without making the walls thicker

Why marks are not awarded

- ✗ "For strength" with no detail

WE DO — Guided practice

Use the starters and word bank to help you

A TV remote control casing is injection moulded in two parts.

2 State two features that show the casing was injection moulded. (2 marks)

Sentence starters

1. …
2. …

Word bank

ejector · Flashing · Split line · injection · sprue · circles · halves · mould

Tip: 2 marks = 2 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) What is the main purpose of a boss on an injection-moulded casing?

☐ Decoration ☐ To take a screw and strengthen the fixing point ☐ To let air escape ☐ To show the recycling code

b) Location pins mainly help with…

☐ Accurate assembly ☐ Corrosion resistance ☐ Aesthetics ☐ Recycling

c) Why are ribs added to the inside of a plastic casing?

☐ To make it heavier ☐ Add rigidity without thick walls and hold internal parts ☐ For grip ☐ To increase cost

d) Thin excess plastic along the edge of a moulded part is called…
☐ Draft ☐ Flashing ☐ Boss ☐ Rib

4 Key terms: write the letter of the matching description next to each term.

1. **Boss**	____	**A** Add strength and rigidity, help material flow, locate internal parts.
2. **Location pin**	____	**B** Raised cylinder to receive a screw; adds strength at the fixing.
3. **Ribs/webs**	____	**C** Thin excess material at a mould's split line.
4. **Flashing**	____	**D** Positions mating parts accurately during assembly.

5 Explain what is meant by ‘Split line’. (2 marks)

6 Explain what is meant by ‘Shear marks’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?